Annex VI. Debt Sustainability Analysis[1]

The public debt has grown rapidly in recent years and is projected to reach about 40 percent of GDP in 2016.[2] The gross financing need is projected to remain high at about 10 percent of GDP in 2016. Over the medium-term fiscal risks remain manageable as the fiscal position is expected to improve. Both the public debt and financing needs will remain below the respective benchmarks even under standard macroeconomic shocks. Assessment of the realism of baseline assumptions for major macroeconomic variables does not point to significant one-sided error, particularly in recent years.

1. **Definition and debt profile.** The public sector debt in this debt sustainability analysis (DSA) includes the obligations of the non-financial public sector (central government and non-financial sector state-owned enterprises), as reported by the authorities, as well as liabilities under advance oil sales. After declining to 17.7 of GDP in 2009, following the 2008 default, the public debt increased to 33.8 percent of GDP in 2015. External debt accounted for about 64 percent of total debt at end-2015, of which about 71 percent was owed to official creditors. Medium- and long-term loans constitute 100 percent of the total public debt stock.

2. **Staff macroeconomic and fiscal assumptions.** Growth is projected at about 1.5 percent and inflation (GDP deflator) at about 0.3 percent in the medium term. The fiscal position is projected to improve with a primary surplus of about 3.1 percent of GDP on average over 2017–21 as the limited financing will restrict the authorities' capacity to carry out their capital expenditure plan. Under these assumptions, the gross financing needs of the public sector are projected to fall to about 4.2 percent of GDP in 2021. After peaking at about 39.6percent of GDP in 2018, the public-debt-to GDP ratio is projected to decline to about 35 percent in 2021. The DSA tool that assesses the realism of projections for real GDP growth, primary balance, and inflation does not indicate significant systematic biases.

3. **Stress tests.** Both the gross financing needs and the public debt-to-GDP ratio remain within the corresponding low-risk thresholds under the standard macroeconomic shocks (i.e., shocks in real GDP growth, primary balance, real interest rate, and exchange rate). The public-debt-to GDP ratio remains below 50 percent and the gross financing needs below 12 percent of GDP even under the combined macro-fiscal shock. Note that as long as the legal limit of 40 percent is binding, corrective fiscal actions are likely to be triggered when debt approaches 40 percent.

4. **Risks and vulnerabilities.** The public DSA risk assessment identifies market perception, and the ratio of public debt held by non-residents as high risks for Ecuador. At about 66 percent of the total debt, non-residents' holding of Ecuador's public debt gives the appearance of being a source of vulnerability, but about 71 percent of the external debt is owed to official creditors and there is limited vulnerability. The high market perception risk is a reflection of the recent hike in spreads

[1] The DSA is based on the baseline scenario presented in the main text and the accompanying tables.

[2] Staff's definition of public debt figures includes liabilities under advance oil sales, which are not considered as public debt in the official definition from the authorities. Public debt excluding liabilities under advance oil sales—which is the relevant definition for the legal ceiling of 40 percent—was about 33 percent of GDP in 2015 and will reach 38.3 percent of GDP in 2018.

©International Monetary Fund. Not for Redistribution

associated with the fall in oil prices. However, near-term rollover risks are limited since there will be no sovereign bond maturing before 2020.

5. **External DSA.** The current account deficit is projected to widen to about 2.3 percent of GDP in 2016, from about ½ a percentage point in 2014, due to lower oil exports (Ecuador's main export commodity) and continuing large fiscal deficits. The expected improvement in the fiscal position will, however, help reduce the current account deficit in the medium term. The current account deficit is expected to be financed mostly by public external borrowing. Consequently, the external debt-to-GDP ratio is projected to increase to about 34.6 percent of GDP in 2018, from 28.5 percent in 2015. The external debt trajectory is sensitive to the non-interest current account shock (e.g. terms of trade shock) and a large (30 percent) real exchange rate depreciation, which is unlikely to materialize due to dollarization.

©International Monetary Fund. Not for Redistribution

Table 1. Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario
(In percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

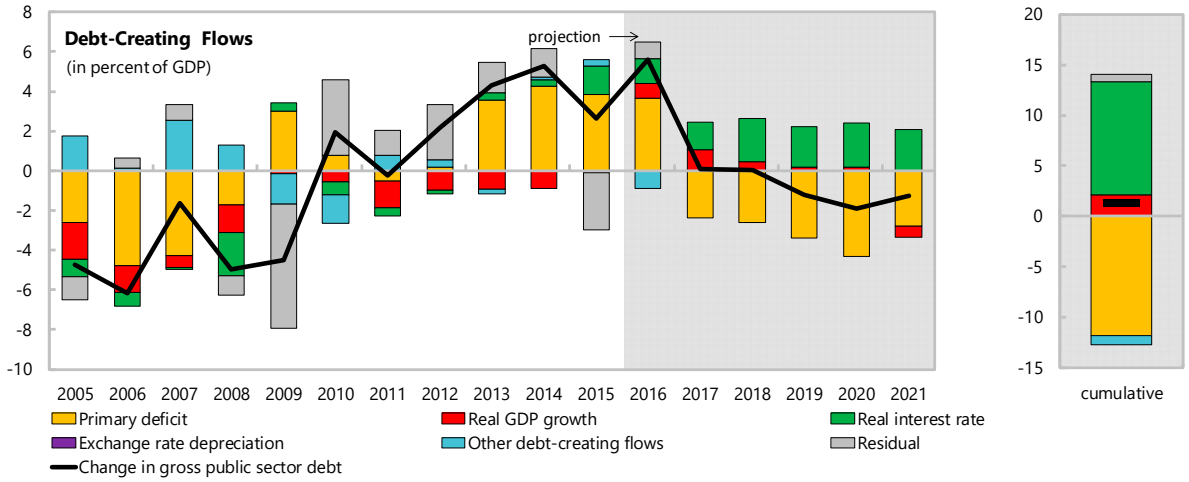
	Actual			Projections					
	2005-2013 [2/]	2014	2015	2016	2017	2018	2019	2020	2021
Nominal gross public debt	24.2	31.2	33.8	39.4	39.5	39.6	38.4	36.5	35.2
Public gross financing needs	3.2	10.2	11.0	10.1	5.0	4.6	4.2	4.3	4.2
Real GDP growth (in percent)	4.5	3.7	0.3	-2.3	-2.7	-1.1	-0.4	-0.5	1.5
Inflation (GDP deflator, in percent)	6.5	2.7	-0.3	0.8	1.8	0.0	0.5	0.1	0.3
Nominal GDP growth (in percent)	11.3	6.5	0.0	-1.5	-0.9	-1.1	0.1	-0.4	1.8
Effective interest rate (in percent) [4/]	4.7	4.2	4.3	4.5	5.3	5.5	5.7	5.8	6.1

As of June 17, 2016

Sovereign Spreads		
EMBIG (bp) 3/		922
5Y CDS (bp)		n.a.
Ratings	Foreign	Local
Moody's	B3	B3
S&Ps	B	B
Fitch	B	B

Contribution to Changes in Public Debt

	Actual			Projections							
	2005-2013	2014	2015	2016	2017	2018	2019	2020	2021	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	-1.5	5.3	2.6	5.6	0.1	0.1	-1.2	-1.9	-1.3	1.4	
Identified debt-creating flows	-1.8	3.8	5.5	4.8	0.1	0.1	-1.2	-1.9	-1.3	0.6	
Primary deficit	-0.7	4.3	3.8	3.6	-2.4	-2.6	-3.4	-4.3	-2.8	-11.8	1.5
Primary (noninterest) revenue and grants	32.1	38.7	33.3	31.8	33.1	32.9	33.3	33.8	34.3	199.1	
Primary (noninterest) expenditure	31.4	42.9	37.1	35.4	30.7	30.3	29.9	29.5	31.5	187.2	
Automatic debt dynamics [5/]	-1.5	-0.6	1.4	2.0	2.5	2.7	2.2	2.4	1.5	13.3	
Interest rate/growth differential [6/]	-1.5	-0.6	1.4	2.0	2.5	2.7	2.2	2.4	1.5	13.3	
Of which: real interest rate	-0.5	0.3	1.5	1.3	1.4	2.2	2.0	2.2	2.1	11.2	
Of which: real GDP growth	-1.0	-0.9	-0.1	0.8	1.1	0.4	0.2	0.2	-0.5	2.1	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.4	0.1	0.3	-0.9	0.0	0.0	0.0	0.0	0.0	-0.9	
Residual, including asset changes [8/]	0.3	1.5	-2.9	0.8	0.0	0.0	0.0	0.0	0.0	0.8	



Source: Fund staff.

1/ Public sector is defined as non-financial public sector.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

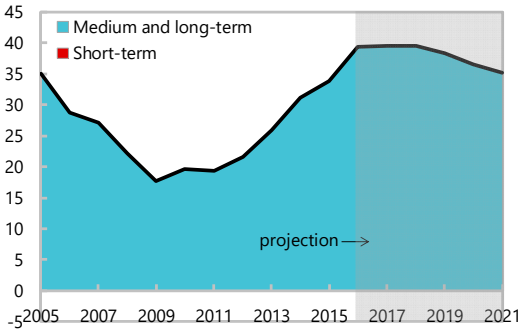
©International Monetary Fund. Not for Redistribution

Figure 1. Ecuador: Public DSA—Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

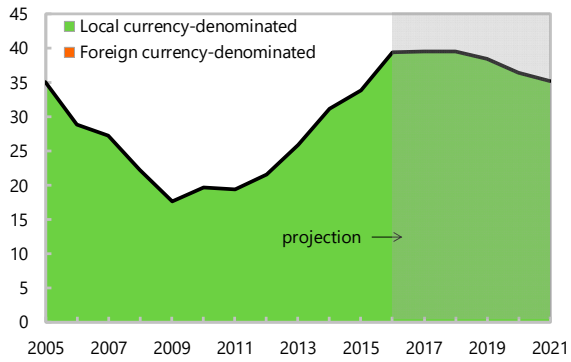
By Maturity

(In percent of GDP)

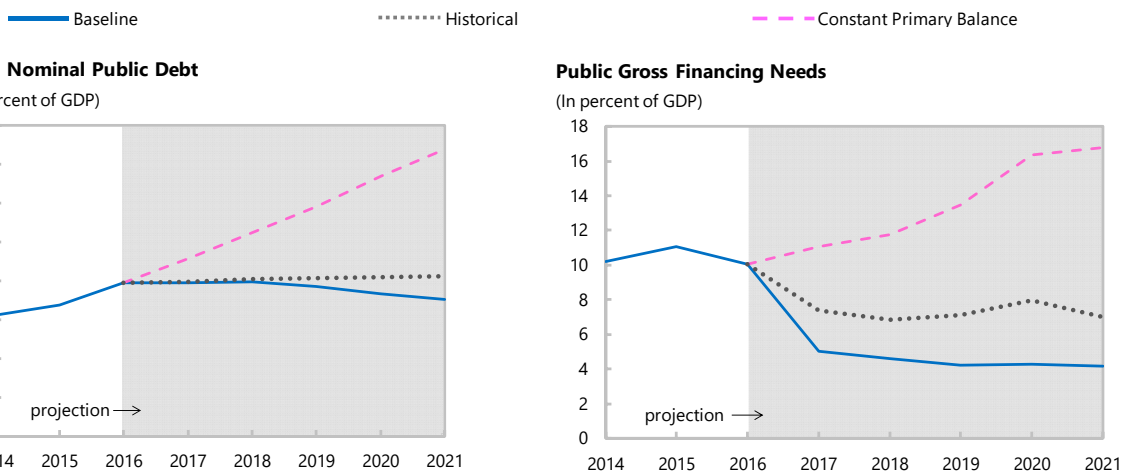


By Currency

(In percent of GDP)



Alternative Scenarios



Underlying Assumptions

(in percent)

Baseline Scenario	2016	2017	2018	2019	2020	2021
Real GDP growth	-2.3	-2.7	-1.1	-0.4	-0.5	1.5
Inflation	0.8	1.8	0.0	0.5	0.1	0.3
Primary Balance	-3.6	2.4	2.6	3.4	4.3	2.8
Effective interest rate	4.5	5.3	5.5	5.7	5.8	6.1
Constant Primary Balance Scenario						
Real GDP growth	-2.3	-2.7	-1.1	-0.4	-0.5	1.5
Inflation	0.8	1.8	0.0	0.5	0.1	0.3
Primary Balance	-3.6	-3.6	-3.6	-3.6	-3.6	-3.6
Effective interest rate	4.5	5.3	5.7	6.1	6.4	6.8

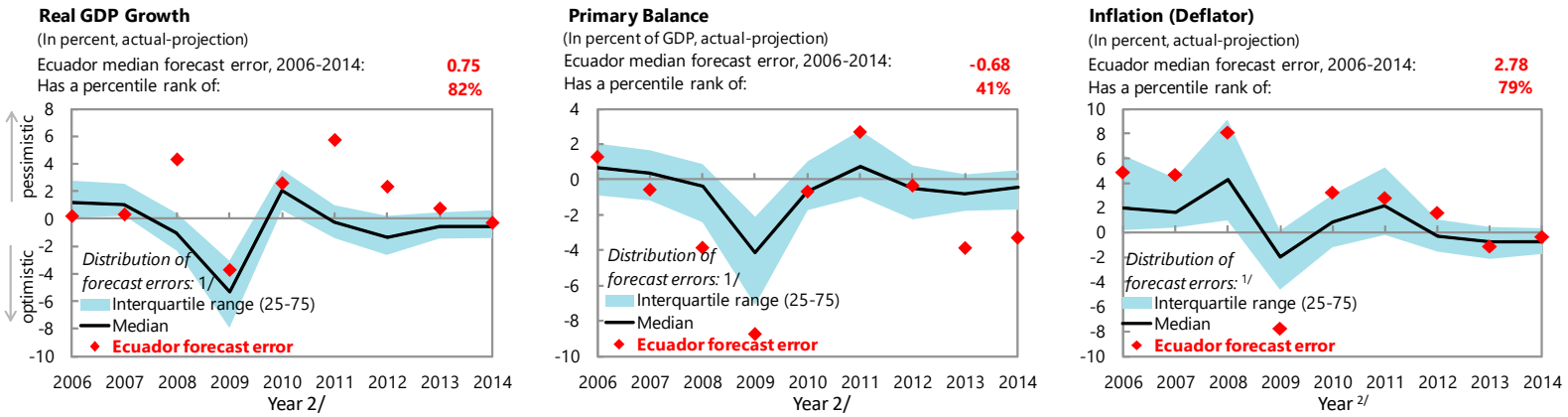
Historical Scenario	2016	2017	2018	2019	2020	2021
Real GDP growth	-2.3	3.9	3.9	3.9	3.9	3.9
Inflation	0.8	1.8	0.0	0.5	0.1	0.3
Primary Balance	-3.6	-0.4	-0.4	-0.4	-0.4	-0.4
Effective interest rate	4.5	5.3	4.6	4.1	3.7	3.4

Source: Fund staff.

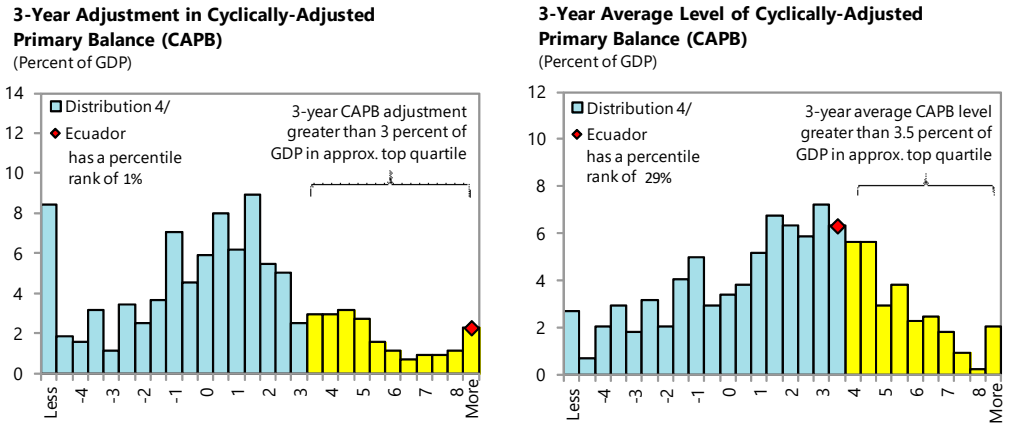
©International Monetary Fund. Not for Redistribution

Figure 2. Ecuador: Public DSA—Realism of Baseline Assumptions

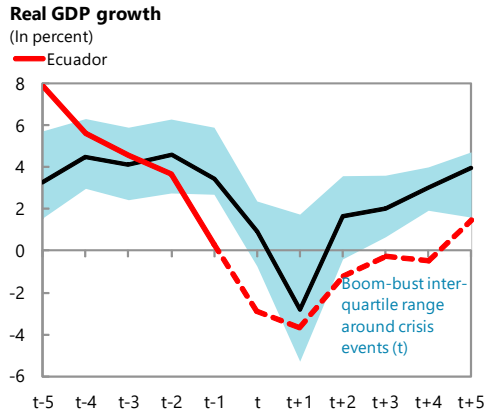
Forecast Track Record, versus all countries



Assessing the Realism of Projected Fiscal Adjustment



Boom-Bust Analysis 3/



Source : Fund Staff.

1/ Plotted distribution includes all countries, percentile rank refers to all countries.

2/ Projections made in the spring WEO vintage of the preceding year.

3/ Ecuador has had a positive output gap for 3 consecutive years, 2013-2015. For Ecuador, t corresponds to 2016; for the distribution, t corresponds to the first year of the crisis.

4/ Data cover annual obervations from 1990 to 2011 for advanced and emerging economies with debt greater than 60 percent of GDP. Percent of sample on vertical axis.

©International Monetary Fund. Not for Redistribution